

STA2453 Final Report Draft

MLSE-PlayerBERT: Player Similarity via Event Sequences

Vision Statement

The goal of this project is to build a player similarity framework that compares soccer players based on **how they play**, rather than only on aggregate outputs such as goals, assists, or pass totals. Using StatsBomb event data and 360 freeze-frame context, the project aims to learn representations of player behavior from ordered sequences of actions. The intended final deliverable is a reproducible modeling pipeline and written report that document the dataset, feature construction, modeling choices, and evaluation strategy for similarity search.

This direction is closely aligned with the collaborator’s goal of producing a practically useful and methodologically defensible player-representation model. We are working toward the same end product: a report and repository that can explain the full workflow from raw data processing to learned player embeddings, and that can support interpretable similarity queries between players.

Introduction

Player comparison in soccer is often based on summary statistics. While these summaries are useful, they do not capture the full process behind player decision-making. Two players may produce similar totals while behaving very differently in context. For example, a winger who repeatedly attempts line-breaking carries and aggressive passes may have similar end-of-match counts to a more conservative wide player, even though their underlying style differs substantially. This motivates a modeling framework that focuses on event sequences and context.

The central idea in this project is that a player’s style can be represented through the sequence of actions they perform during a match. Instead of treating a player as a static row of features, the project represents a player through ordered event embeddings built from on-ball actions and their surrounding game state. This framing naturally leads to a Transformer-based architecture, since Transformers are designed to learn from structured sequences while preserving contextual relationships.

The working objective is therefore to learn one embedding per player that can be used for similarity search. A good embedding should place players with similar playing styles close together, while separating players whose event patterns, tactical roles, and contextual decisions differ.

Collaborator Alignment

The project is being developed as a shared deliverable with a collaborator, so alignment has been important from the beginning. At this stage, our goals are

aligned in three ways.

First, we agree on the conceptual goal: the project should measure player style from event process rather than from outcome totals alone. Second, we agree on the technical goal: the final repository should contain a reproducible end-to-end pipeline, including preprocessing, model training, and inference. Third, we agree on the deliverable goal: by the end of term, the work should be mature enough to hand off clearly, with all code, outputs, and documentation organized for submission and collaborator review.

This working draft is meant to support that alignment by making the project’s purpose, data, and current approach explicit before the final evaluation and conclusions are written.

Data

The project uses the **StatsBomb Open Data** repository as its primary data source. The two main components used so far are:

1. **StatsBomb event data**, which records event-level match actions such as passes, dribbles, carries, shots, duels, blocks, and clearances.
2. **StatsBomb 360 data**, which provides freeze-frame context around an event, including the visible positions of surrounding players and contextual spatial information.

These two sources are joined so that each modeled event contains both the event attributes themselves and the spatial context around the event. In the current workflow, the combined dataset is processed into a flattened JSONL format. Nested dictionaries are converted into dot-separated features, while context-rich list structures such as `freeze_frame` are retained for separate handling. The preprocessing pipeline also removes unique identifiers, standardizes missing values, normalizes categorical representations, and drops rare features with extremely low coverage.

The resulting working dataset contains **over one million event-plus-context records**, derived from StatsBomb event and 360 data. At the modeling stage, the relevant unit is not only the single event, but also the **ordered event sequence for a player within a match**. This means the data are used at two levels:

1. **Event level**: one event plus its local context.
2. **Sequence level**: one player’s ordered series of events within a match.

This structure is central to the design of the current modeling framework.

Methods

Preprocessing and Feature Construction

The first major component of the project is data preprocessing. The raw event and 360 records are joined and transformed into a model-ready representation. Several preprocessing choices have been implemented so far:

1. Removal of identifiers that are unique to individual events and therefore not useful as generalizable predictive features.
2. Standardization of missing values for boolean and categorical variables.
3. Flattening of nested event dictionaries into dot-key features.
4. Preservation of freeze-frame information for separate contextual encoding.
5. Bucketization or normalization of certain numerical/spatial variables so they can be incorporated more consistently into model inputs.

This preprocessing step produces a structured event representation that is suitable for both feature-token encoding and contextual scene modeling.

Baseline EventEncoder

The current baseline model begins with an **EventEncoder**, whose purpose is to convert a single event and its 360 context into a dense fixed-dimensional embedding. In the baseline version, event attributes are treated as tokens, where each token corresponds to a feature-value pair. Feature-specific vocabularies are built from the processed dataset, and token embeddings are learned jointly with feature identity embeddings.

The tokenized event attributes are passed through a Transformer encoder to produce an event representation. In parallel, the freeze-frame context is encoded by transforming visible surrounding players into vectors based on their relative positions and role indicators such as teammate and keeper flags. These player-level vectors are pooled into a frame-level representation. The event and frame representations are then fused through a learned gate to produce the final event embedding.

This baseline design allows the model to represent both the event itself and the local tactical environment in which it occurred.

PlayerBERT

The second major component is **PlayerBERT**, a sequence model that operates on ordered event embeddings. The core idea is that a player’s style is better represented by the sequence of actions they perform than by any single isolated event. In the current setup, each player’s events are ordered within a match using period, minute, second, timestamp, and event index. These ordered event embeddings are then passed into an encoder-only Transformer with positional embeddings.

The baseline self-supervised objective is **masked event modeling**. A subset of event embeddings in the sequence is masked, and the model is trained to reconstruct the original event embeddings at the masked positions. This allows the sequence model to learn context-sensitive representations of player behavior without requiring external labels.

At the moment, the baseline PlayerBERT training uses EventEncoder outputs as initialization and keeps the EventEncoder frozen for simplicity. One of the main planned improvements is to move toward staged or end-to-end fine-tuning so that event-level and sequence-level objectives can be optimized jointly.

Current V2 Extension

In addition to the baseline pipeline, a stronger V2 event encoder has already been drafted in the repository. This version introduces:

1. Event-conditioned scene encoding through a learned query token.
2. Relational self-attention over freeze-frame players with geometric bias.
3. Cross-attention fusion between event tokens and scene context.
4. Multi-task self-supervised learning, including masked attribute modeling, masked player-token reconstruction, and a contrastive objective.

This V2 design is intended to address a key limitation of the baseline model: simple pooling may not capture the relational structure of the 360 scene well enough. By explicitly modeling player-to-player geometry and event-conditioned context, the V2 encoder is expected to produce more discriminative event representations.

Progress So Far

The project has moved past the initial setup stage and now has a functioning baseline pipeline. So far, the following pieces are in place:

1. A preprocessing script that joins event and 360 data.
2. A data processing notebook that cleans, normalizes, flattens, and filters features.
3. A baseline EventEncoder training workflow.
4. A PlayerBERT training workflow for sequence modeling.
5. An inference workflow that generates player embeddings and supports nearest-neighbor similarity search.
6. A draft V2 event encoder and training scaffold for the next round of experiments.

Initial training runs confirm that the baseline pipeline is working end to end. The EventEncoder training loss decreases across epochs, and the PlayerBERT pipeline successfully produces saved model checkpoints and player embedding outputs. Inference has already been tested for multiple example players, which confirms that the retrieval side of the workflow is operational. At this stage, however, these outputs should be treated as preliminary and not as final results.

Planned Evaluation

The final version of the project will require stronger evaluation than qualitative nearest-neighbor examples alone. The current plan is to evaluate the learned embeddings in two complementary ways.

First, the project will include **qualitative case studies**, where similarity outputs for well-known players are examined to determine whether retrieved neighbors make sense in terms of role and playing style. Second, the project will incorporate **more formal retrieval diagnostics**, such as checking whether players with similar positions, tactical roles, or action profiles are placed close together in embedding space.

Because this draft is intended as a methods-focused working version of the final report, those final evaluation results are intentionally not included yet.

Discussion and Next Steps

The current state of the project shows that the main conceptual and technical framework is in place. The dataset has been assembled and processed, the baseline event and sequence models have been implemented, and the inference pipeline is functional. The next phase is to strengthen the model and the report in parallel.

The most important next steps are:

1. Train and compare the V2 event encoder against the baseline version.
2. Improve evaluation beyond simple nearest-neighbor inspection.
3. Test whether end-to-end tuning improves player-level embeddings.
4. Develop clearer figures that explain the data pipeline, model architecture, and embedding workflow.
5. Expand the report with final experimental comparisons, limitations, and interpretation.

In its final form, the project should provide both a technically sound machine learning pipeline and a clear statistical narrative: why event sequences and context matter, how the data were structured, what modeling assumptions were made, and how the learned player representations should be interpreted.

Conclusion

This draft outlines the core structure of the final deliverable. The project aims to learn player similarity from event sequences and 360 contextual data using a Transformer-based architecture. The collaborator and I are aligned on producing a reproducible workflow and a report that clearly explains the data, the methods, and the reasoning behind the modeling choices. At this stage, the strongest parts of the work are the data pipeline, the baseline modeling framework, and the initial retrieval workflow. Final results, quantitative evaluation, and polished figures will be added after the next round of experiments.